

Parametric Equations



Derivatives of parametric equations

- 1 For $x(t) = t^2$, $y(t) = t^3$, find $\frac{dy}{dx}$ in terms of t .
 - 2 For $x(t) = \cos t$, $y(t) = \sin t$, find $\frac{dy}{dx}$ and evaluate it at $t = \frac{\pi}{4}$.
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Second derivatives of parametric equations

- 3 For $x(t) = t^2$, $y(t) = t^3$, find $\frac{d^2y}{dx^2}$ using $\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}(\frac{dy}{dx})}{\frac{dx}{dt}}$.
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Tangent lines to parametric curves

- 4 Find the equation of the tangent line to the curve $x(t) = t^2 - 1$, $y(t) = t^3 - t$ at $t = 2$.
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Arc length of parametric curves

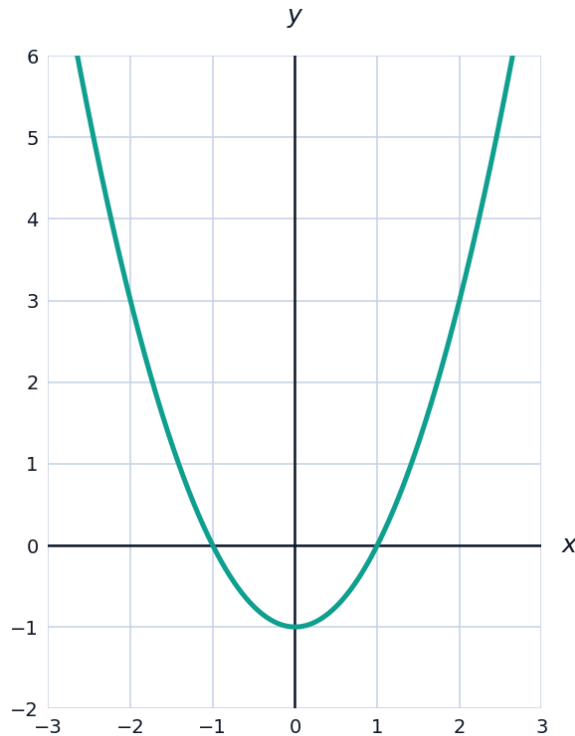
- 5 State the arc length formula for a parametric curve $x(t)$, $y(t)$ on $[a, b]$.
 - 6 Find the arc length of $x(t) = 3t$, $y(t) = 4t$ for $t \in [0, 2]$.
 - 7 Set up (but do not evaluate) the arc length integral for $x(t) = t^2$, $y(t) = t^3$ on $[0, 1]$.
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Speed as a vector quantity

- 8 A particle moves with position $x(t) = t^2$, $y(t) = 2t$. Find its speed at $t = 1$.
 - 9 A particle moves with position $x(t) = 3\cos t$, $y(t) = 3\sin t$. Find its speed at any time t , and interpret the result.
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Eliminating the parameter

- 10 Eliminate the parameter from $x(t) = t$, $y(t) = t^2 - 1$ to find the Cartesian equation, and sketch it.



- 11 Eliminate the parameter from $x(t) = 2t + 1$, $y(t) = 4t - 3$ to find the Cartesian equation.
- 12 Explain why eliminating the parameter from $x(t) = \cos t$, $y(t) = \sin t$ using the identity $\sin^2 t + \cos^2 t = 1$ gives a relation that is NOT a function $y = f(x)$.

Concavity of parametric curves

- 13 For $x(t) = t^2$, $y(t) = t^3$ (with $\frac{d^2y}{dx^2} = \frac{3}{4t}$ found earlier), determine the concavity of the curve for $t > 0$ and for $t < 0$.

Horizontal and vertical tangents

- 14 For $x(t) = t^2 - 4$, $y(t) = t^3 - 3t$, find the value(s) of t where the tangent line is horizontal (where $\frac{dy}{dt} = 0$, provided $\frac{dx}{dt} \neq 0$).
- 15 For the same curve, find the value(s) of t where the tangent line is vertical (where $\frac{dx}{dt} = 0$, provided $\frac{dy}{dt} \neq 0$).

Motion along a parametric path (particle interpretation)

- 16 A particle's position is $x(t) = t^2$, $y(t) = t^3 - 3t$ for $t \in [-2, 2]$. Find all times when the particle is moving straight up or down (vertical velocity component only, i.e. $\frac{dx}{dt} = 0$).
- 17 For the particle above, find the speed at $t = 2$.
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Arc length of a full circle (verifying the circumference formula)

- 18 Use the parametric arc length formula to verify that the circumference of a circle $x(t) = r \cos t$, $y(t) = r \sin t$ for $t \in [0, 2\pi]$ is $2\pi r$.
- 19 Find the arc length of the parametric curve $x(t) = e^t \cos t$, $y(t) = e^t \sin t$ is a spiral; set up (but do not evaluate) its arc length integral on $[0, \pi]$.
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Parametric equations from real motion

- 20 A Ferris wheel of radius 10 m completes one revolution every 40 s, centered 12 m above the ground. Write parametric equations for a rider's height above the ground as a function of time t (seconds), assuming the rider starts at the bottom.
- 21 For the Ferris wheel model above, find the rider's vertical velocity $\frac{dy}{dt}$ at $t = 10$ s (a quarter revolution).